Tightly-Coupled RAM (TCRAM) Module

This chapter describes the tightly-coupled RAM (TCRAM) module.

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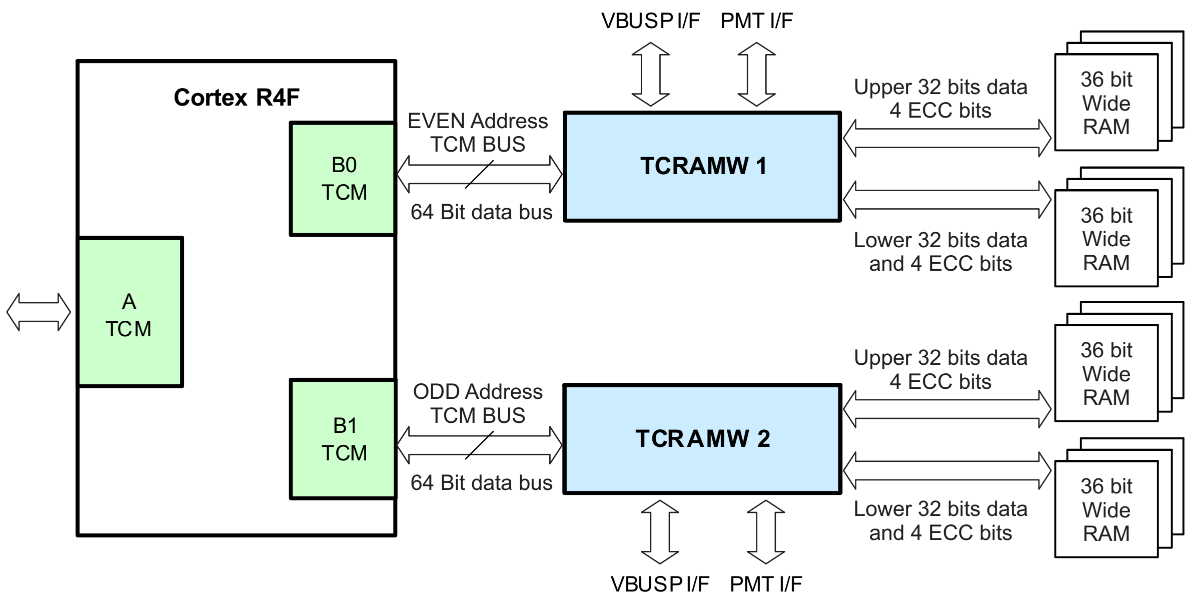
6.1 Overview

The Hercules family of microcontrollers are based on the ARM Cortex-R4F processor. This CPU has two tightly-coupled memory interfaces – ATCM and BTCM, which are used to interface to the program and data memories, respectively. The Hercules MCUs use the ATCM interface for the main flash memory and the BTCM interface for the CPU data RAM.

6.1.1 B0TCM and B1TCM Connection Diagram

The BTCM interface is further divided into two parts – B0TCM and B1TCM, which are both used to interface to actual RAM banks as shown in Figure 6-1.

Figure 6-1. TCRAM Module Connections



6.1.2 Main Features

The main features of the tightly-coupled RAM interface module are:

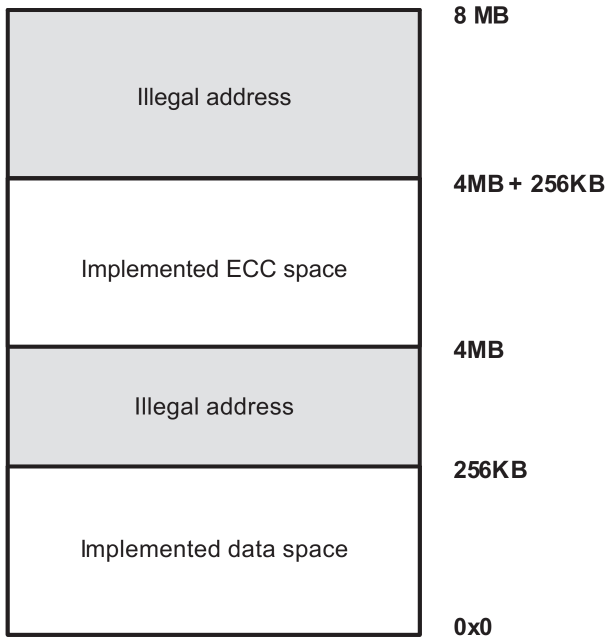
- Controls read/write accesses to the data RAM
- Decodes addresses within the memory region allocated for the RAM
- Supports read and write accesses in 64-bit, 32-bit, 16-bit or 8-bit access sizes
  - Does not support bit-wise operations
- Safety Features:
  - Support for Cortex-R4F CPU's Built-In Single-Error-Correction Double-Error-Detection (SECDED) Logic
    - Uses the CPU's Event bus and maintains the SECDED status in memory-mapped registers
    - Captures the number of occurrences of single-bit or multi-bit errors as well as the RAM address that has the fault
    - Generates signals for indicating single-bit and multi-bit errors to the Error Signaling Module (ESM)
  - Support for Cortex-R4F CPU's Parity Protection Logic for BTCM Address Bus and Control Signals
    - Uses the CPU's TCM Address Parity Scheme and indicates an address bus parity error to the ESM

- Redundant Address Decode Scheme
  - Checks the decoding of CPU address lines and generation of correct memory selects for the RAM banks
  - Also supports checking of the redundant address decode comparators themselves
- Supports auto-initialization of the CPU data RAM banks

6.2 RAM Memory Map

The ARM Cortex-R4F CPU allows up to 8MB to be accessed through the BTCM interface. The Hercules family of microcontrollers support up to 256KB RAM on the BTCM interface. Check the specific part's datasheet to identify the actual amount of TCRAM supported on the part. This RAM is protected by ECC allowing the CPU to correct any single-bit errors and detect double-bit errors within a 64-bit value. The error correction codes (ECC) are stored in the RAM memory space as well. The memory map for the TCRAM and the corresponding ECC space is shown in Figure 6-2. Any access to an unimplemented TCRAM location results in an error response from the TCRAM module.

Figure 6-2. RAM Memory Map



Each RAM data word is 64 bits wide. These 64 bits are divided into two 32 bits per RAM bank as shown in Figure 6-1. The 8 bits of ECC are also divided into 4 bits per RAM bank.

For every 64-bit read from the RAM, an 8-bit ECC is also read by the CPU on its ECC bus. Similarly, for every 64-bit write to the RAM, the CPU also writes an 8-bit ECC using the same ECC bus.

NOTE: **Read-Modify-Write Requirement for Writes to RAM:** The TCRAM interface module supports 64-bit, 32-bit, 16-bit or 8-bit writes to the RAM. However the ECC is calculated by the CPU for 64-bit values only. For any write access smaller than 64 bits, it is necessary to force the CPU to perform a 64-bit read-modify-write operation in order to ensure that the correct ECC is also written. This can be done by setting the bit 1: BTCMRMW of c15, the Secondary Auxiliary Control Register of the CPU. **This bit is already set by default.**

The ECC memory can also be directly accessed via memory-mapped offset addresses starting from 4MB, as shown in Figure 6-2. A read from the ECC space results in the 8-bit ECC value appearing on each byte of the 64-bit CPU data. The ECC memory can only be written to as a 64-bit access. The write to the ECC space must also first be enabled via the RAM Control Register (RAMCTRL).

NOTE: No ECC Error Generated for Accesses to ECC Memory: A read from the ECC space send the ECC value on both the 64-bit TCM read data bus as well as the 8-bit ECC bus. This could result in the detection of a multi-bit error by the SECDED logic inside the CPU. The TCRAM interface module ignores the ECC error indicated by the CPU for the access to the ECC space.

6.3 Safety Features

The TCRAM interface module incorporates some features that are designed specifically with safety considerations. These are described in the following sections.

6.3.1 Support for Cortex-R4F CPU's Single-Error-Correction Double-Error-Detection (SECDED)

The TCRAM interface module monitors the CPU's event bus. The CPU's event bus signals single-bit or multi-bit errors for B0TCM as well as B1TCM separately. These signals are monitored by the TCRAM modules for each of these interfaces.

TCRAM Interface Module Features dedicated for SECDED Support:

- Dedicated single-bit error counter
  - This counter is stored in a memory-mapped register called RAMOCCUR
  - RAMOCCUR is used to count the single-bit errors corrected by the CPU's SECDED logic
  - The TCRAM interface module allows the application to generate an interrupt via the RAMINTCTRL register when the number of single-bit errors corrected by the CPU exceeds a programmable threshold, RAMTHRESHOLD
- RAM Error Status Register
  - The errors detected by the TCRAM interface module as well as those indicated by the CPU are flagged in the RAMERRSTATUS register
  - There are separate bits to indicate single-bit error, double-bit error, address decode failure, address compare logic failure, read-address parity failure, and write-address parity failure
- ECC Error Address Capture
  - Separate registers to hold the address on which a single-bit error is detected (RAMSERRADDR) or a double-bit error is detected (RAMUERRADDR)
  - The RAMSERRADDR register is only updated when the RAMTHRESHOLD value is set to 1
  - Both the RAMSERRADDR and RAMUERRADDR capture the 64-bit-aligned address for the access to the TCRAM as an offset from the base address of the TCRAM (0x08000000 by default)

NOTE: Cortex-R4F CPU Event Bus Signaling Not Enabled By Default: Upon power-up and after a CPU reset, the event signaling mechanism inside the Cortex-R4F CPU is disabled. This feature must be enabled by setting the Export (X-bit) of the Performance Monitoring and Control Register (PMNC) in the CPU. The TCRAM interface module can only capture the single-bit or double-bit ECC error occurrences once the CPU's event signaling mechanism is enabled.

6.3.2 Support for Cortex-R4F CPU's Address and Control Bus Parity Checking

The Cortex-R4F CPU calculates a single parity-bit for the TCRAM address and control signals. The TCRAM interface module also computes this parity bit based on the CPU's address bus and control signals. The computed parity bit is compared against the parity bit received from the CPU. A mismatch is signaled as an Address Parity Failure to the Error Signaling Module (ESM) group2 channel 10 or 12. There is a separate address parity failure error channel for B0TCM and B1TCM.

The 64-bit TCRAM address which fails the parity check is captured in the RAMPERRADDR register as an offset from the base address of the TCRAM (0x08000000 by default). The TCRAM interface module also indicates the type of access, read or write, that failed the parity check. This is indicated by the RADDR_PAR_FAIL or the WADDR_PAR_FAIL status flags in the RAMERRSTATUS register.

The RAMERRSTATUS and RAMPERRADDR registers must be cleared by the application in order for the TCRAM interface module to continue capturing subsequent errors and error addresses.

The parity scheme used for the described parity checking mechanism is defined by the global system parity selection. This can be configured using the DEVPARSEL field of the DEVCR1 control register in the system module. This device-wide parity scheme can be overridden inside the TCRAM interface module by configuring the Address Parity Override field in the RAMCTRL register.

NOTE: **No Change Of Parity Scheme On-The-Fly:** The TCRAM interface module does not support on-the-fly change to the parity scheme being used for checking the CPU address bus and control bus. The application must ensure that the parity polarity (odd or even) is not changed while there is an ongoing access to the TCRAM.

6.3.3 Redundant Address Decode

The TCRAM interface module generates the memory selects for each of the TCRAM banks as well as the ECC memory based on the CPU address. The logic to generate these memory selects is duplicated and the outputs compared to detect any address decode errors. A mismatch is indciated as an Address Error to the Error Signaling Module (ESM), one signal for B0TCM and one for B1TCM. The TCRAM or ECC address that caused the fault is captured in the RAMUERRADDR register. This 64-bit-aligned address is stored as an offset from the base of the TCRAM or ECC memory.

As described earlier, each individual physical RAM bank is 36 bits wide. Each RAM bank contributes 32 bits of data and 4 bits of ECC when the bus master performs a 64-bit read from the TCRAM. Each TCRAM bank receives a memory select and the address from the TCRAM interface module. Any difference between the address and the memory selects results in wrong data and ECC pair being sent to the CPU. The CPU's SECDED block will detect this data error.

The TCRAM interface module also supports a mechanism to test the operation of the redundant address decode logic and the compare logic. This testing is supported by providing a test stimulus, and can be triggered by the application by configuring the RAMTEST register. The address of any error identified during testing of the redundant address decode and compare logic is not captured in the RAMUERRADDR register.

NOTE: **Address decode checking when in compare logic test mode:** When the address decode and compare logic test mode is enabled, the redundant address decode and compare logic is not available for checking the proper generation of the memory selects for the TCRAM and ECC memory.

6.4 TCRAM Auto-Initialization

The RAM memory can be initialized by using the dedicated auto-initialization hardware. The TCRAM Module initializes the entire memory when the auto-initialization is enabled by the INIT_DOMAIN register upon receiving a MMI_INIT pulse from the system module. All enabled RAM data memory locations are initialized to zeros and the ECC memory is initialized to the correct ECC value for zeros, that is, 0Ch.

6.5 Trace Module Support

No data is traced for an access to ECC memory.

6.6 Emulation / Debug Mode Behavior

The following describes the behavior of the TCRAM Module when in debug mode:

- The RAMOCCUR register continues to count the single-bit error corrections performed by the Cortex-R4F CPU's SECDED logic.
- No single-bit error interrupt is generated nor is any single-bit error address captured even when the RAMOCCUR counter reaches the programmed single-bit error correction threshold.
- No uncorrectable error interrupt is generated nor is any double-bit error address captured.
- No address parity error interrupt is generated nor is any parity error address captured.
- The RAMUERRADDR register is not cleared by a read in debug mode.
  – That is, if a double-bit error address is captured and is not read by the CPU before entering debug mode, then it remains frozen during debug mode even if it is read.
- The RAMPERRADDR register is not cleared by a read in debug mode.

6.7 TCRAM_MODULE_CONTROL_AND_STATUS Registers

Table 6-1 lists the memory-mapped registers for the TCRAM_MODULE_CONTROL_AND_STATUS. All register offset addresses not listed in Table 6-1 should be considered as reserved locations and the register contents should not be modified.

The TCRAM Module registers are accessed through the system module registers' space in the Cortex-R4F CPU's memory map. All registers are 32-bit wide and are located on a 32-bit boundary. Reads and writes to registers are supported in 8-, 16-, and 32-bit accesses. The base address for the control registers is FFFF F800h for even RAM ECC and FFFF F900h for odd RAM ECC.

Table 6-1. TCRAM_MODULE_CONTROL_AND_STATUS Registers

Offset	Acronym	Register Name	Section
0h	RAMCTRL	TCRAM Module Control Register	Section 6.7.1
4h	RAMTHRESHOLD	TCRAM Module Single-Bit Error Correction Threshold Register	Section 6.7.2
8h	RAMOCCUR	TCRAM Module Single-Bit Error Occurrences Control Register	Section 6.7.3
Ch	RAMINTCTRL	TCRAM Module Interrupt Control Register	Section 6.7.4
10h	RAMERRSTATUS	TCRAM Module Error Status Register	Section 6.7.5
14h	RAMSERRADDR	TCRAM Module Single-Bit Error Address Register	Section 6.7.6
1Ch	RAMUERRADDR	TCRAM Module Uncorrectable Error Address Register	Section 6.7.7
30h	RAMTEST	TCRAM Module Test Mode Control Register	Section 6.7.8
38h	RAMADDRDECVECT	TCRAM Module Test Mode Vector Register	Section 6.7.9
3Ch	RAMPERRADDR	TCRAM Module Parity Error Address Register	Section 6.7.10
40h	INIT_DOMAIN	Auto-Memory Initialization Enable Register	Section 6.7.11

Complex bit access types are encoded to fit into small table cells. Table 6-2 shows the codes that are used for access types in this section.

Table 6-2. TCRAM_MODULE_CONTROL_AND_STATUS Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W1CP	1C P W	1 to clear Requires privileged access Write
WP	P W	Requires privileged access Write
Reset or Default Value		
-n		Value after reset or the default value

6.7.1 RAMCTRL Register (Offset = 0h) [reset = 0005000Ah]

RAMCTRL is shown in Figure 6-3 and described in Table 6-3.

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RAMCTRL controls the safety features supported by the TCRAM Module.

Figure 6-3. RAMCTRL Register

31	30	29	28	27	26	25	24
RESERVED				ADDR_PARITY_OVERRIDE			
R-0h				R/WP-0h			
23	**22**	**21**	**20**	**19**	**18**	**17**	**16**
RESERVED				ADDR_PARITY_DISABLE			
R-0h				R/WP-5h			
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**
RESERVED							ECC_WR_EN
R-0h							R/WP-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
RESERVED				ECC_DETECT_EN			
R-0h				R/WP-Ah			

Table 6-3. RAMCTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-28	RESERVED	R	0h	Reads return 0. Writes have no effect.
27-24	ADDR_PARITY_OVERRIDE	R/WP	0h	Address Parity Override. This field, when set to Dh, will invert the parity scheme selected by the device global parity selection. The address parity checker would then work on the inverted parity scheme. By default, the parity scheme is the same as the global device parity scheme. All other values = Parity scheme is the same as the device global parity scheme. Dh = Parity scheme is opposite to the device global parity scheme.
23-20	RESERVED	R	0h	Reads return 0. Writes have no effect.
19-16	ADDR_PARITY_DISABLE	R/WP	5h	Address Parity Detect Disable. This field, when set to Ah, disables the parity checking for the address bus. The parity checking is enabled when this field is set to any other value. **Note: The application must ensure that the WADDR_PAR_FAIL and RADDR_PAR_FAIL bits in RAMERRSTATUS register are cleared before enabling address parity checking.** All other values = Address parity checking is enabled. Ah = Address parity checking is disabled.
15-9	RESERVED	R	0h	Reads return 0. Writes have no effect.
8	ECC_WR_EN	R/WP	0h	ECC Memory Write Enable. This bit is provided to prevent accidental writes to the ECC memory. A write access to the ECC memory is allowed only when the ECC_WR_EN bit is set to 1. If this bit is cleared, then any writes to ECC memory are ignored. **Note: Reads are allowed from the ECC memory regardless of the state of the ECC_WR_EN bit.** 0h = ECC memory writes are disabled. 1h = ECC memory writes are enabled.
7-4	RESERVED	R	0h	Reads return 0. Writes have no effect.

Table 6-3. RAMCTRL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3-0	ECC_DETECT_EN	R/WP	Ah	ECC Detect Enable. This is a 4-bit key to enable the ECC detection feature in the TCRAM Module. If this field is set to any value other than 5h, then the TCRAM Module starts monitoring the TCM event bus and generates the corresponding error status flags. The error status updates are done only when the ECC_DETECT_EN field is not 5h. The ECC detection is enabled by default, as the ECC_DETECT_EN field default value is Ah. All other values = ECC detection is enabled. 5h = ECC detection is disabled.

6.7.2 RAMTHRESHOLD Register (Offset = 4h) [reset = 0h]

RAMTHRESHOLD is shown in Figure 6-4 and described in Table 6-4.

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RAMTHRESHOLD allows the application to configure the number of single-bit error corrections by the SECDED logic inside the Cortex-R4F CPU before generating a single-bit error interrupt.

Figure 6-4. RAMTHRESHOLD Register

31-16	15-0
RESERVED	THRESHOLD
R-0h	R/WP-0h

Table 6-4. RAMTHRESHOLD Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0. Writes have no effect.
15-0	THRESHOLD	R/WP	0h	Single-bit Error Threshold Count. This field contains the threshold value for the Single-bit Error Correction (SEC) occurrences before the single-bit error interrupt is generated. If this threshold is set to 1 then all single-bit error addresses are captured. To enable the error occurrence detection, the threshold must be set to a non-zero value.

6.7.3 RAMOCCUR Register (Offset = 8h) [reset = 0h]

RAMOCCUR is shown in Figure 6-5 and described in Table 6-5.

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RAMOCCUR indicates the number of single-bit error corrections performed by the SEC logic inside the Cortex-R4F CPU.

Figure 6-5. RAMOCCUR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
SINGLE_ERROR_OCCURRENCES															
R/WP-0h															

Table 6-5. RAMOCCUR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0. Writes have no effect.
15-0	SINGLE_ERROR_OCCURRENCES	R/WP	0h	Single-bit Error Correction Occurrences. This 16-bit counter contains the number of single-bit error occurrences. RAMOCCUR is reset to zero when it becomes equal to the THRESHOLD value set in the RAMTHRESHOLD register. The application must clear the RAMOCCUR register by writing 0x0 before setting the THRESHOLD value. If the RAMOCCUR value is already higher than the programmed THRESHOLD value, then the counter increments and wraps around (overflow) to zero. **Note: If the application tries to clear the RAMOCCUR register at the same time as the TCRAM Module tried to update it, then the TCRAM Module takes priority.** **Note: When the RAMTHRESHOLD register is set to 1, then the RAMOCCUR register must be cleared whenever a single-bit error correction occurs in order to count subsequent single-bit error corrections.**

6.7.4 RAMINTCTRL Register (Offset = Ch) [reset = 0h]

RAMINTCTRL is shown in Figure 6-6 and described in Table 6-6.

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RAMINTCTRL enables the generation of an interrupt to the CPU whenever the number of single-bit error corrections (RAMOCCUR) reaches the programmed threshold (RAMTHRESHOLD).

Figure 6-6. RAMINTCTRL Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							SERR_EN
R-0h							R/WP-0h

Table 6-6. RAMINTCTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0. Writes have no effect.
0	SERR_EN	R/WP	0h	Single-bit Error Correction Interrupt Enable. This bit, when set to 1, enables the generation of the single-bit error interrupt when the RAMOCCUR count reaches the programmed RAMTHRESHOLD. If the interrupt is not enabled, the single-bit error counter continues to count by resetting back to zero without generating any error interrupt. The SERR status flag in the RAMERRSTATUS register gets set regardless of whether the SERR interrupt is enabled or not. 0h = Single-bit error generation is disabled. 1h = Single-bit error generation is enabled.

6.7.5 RAMERRSTATUS Register (Offset = 10h) [reset = 0h]

RAMERRSTATUS is shown in Figure 6-7 and described in Table 6-7.

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RAMERRSTATUS indicates the status of the various error conditions monitored by the TCRAM Module.
For equality test:

If the comparator matches (no true silicon fail), there is no status bit set for ADDR_COMP_LOGIC_FAIL or ADDR_DEC_FAIL.

If there is true silicon malfunction, ADDR_COMP_LOGIC_FAIL and ADDR_DEC_FAIL will be set, no UERRADDRESS is captured.

For inequality test, the compare vector will not match since non-inverted and inverted values of the same test vector are fed to the comparator:

If there is no silicon malfunction on any of the comparator bits, then only ADDR_DEC_FAIL will be set. This is chosen so that we can ensure the functional ADDR_DEC_FAIL status bit data path can be tested.

If there is a silicon malfunction on any of the comparator bits, then, ADDR_COMP_LOGIC_FAIL and ADDR_DEC_FAIL will be set, no UERRADDRESS is captured.

Figure 6-7. RAMERRSTATUS Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	**22**	**21**	**20**	**19**	**18**	**17**	**16**
RESERVED							
R-0h							
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**
RESERVED						WADDR_PAR_FAIL	RADDR_PAR_FAIL
R-0h						R/W1CP-0h	R/W1CP-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
RESERVED		DERR	ADDR_COMP_LOGIC_FAIL	RESERVED	ADDR_DEC_FAIL	RESERVED	SERR
R-0h		R/W1CP-0h	R/W1CP-0h	R-0h	R/W1CP-0h	R-0h	R/W1CP-0h

Table 6-7. RAMERRSTATUS Register Field Descriptions

Bit	Field	Type	Reset	Description
31-10	RESERVED	R	0h	Reads return 0. Writes have no effect.
9	WADDR_PAR_FAIL	R/W1CP	0h	This bit indicates a Write Address Parity Failure. This bit must be cleared by writing 1 to it in order to enable the capture of parity error address for subsequent failures. This bit must be in a cleared state for generation of any new parity error interrupt.
8	RADDR_PAR_FAIL	R/W1CP	0h	This bit indicates a Read Address Parity Failure. This bit must be cleared by writing 1 to it in order to enable the capture of parity error address for subsequent failures. This bit must be in a cleared state for generation of any new parity error interrupt.
7-6	RESERVED	R	0h	Reads return 0. Writes have no effect.
5	DERR	R/W1CP	0h	This bit indicates a multi-bit error detected by the Cortex-R4F SECDED logic.
4	ADDR_COMP_LOGIC_FAIL	R/W1CP	0h	Address decode logic element failed. This bit indicates that the redundant address decode logic test scheme has detected that a compare element has malfunctioned during the testing of the logic. This bit has to be cleared by writing 1 to it in order to enable the capture of uncorrectable error address for subsequent failures. This bit has to be in a cleared state for generation of a new uncorrectable error interrupt. This bit only gets set in the test mode, and has no relevance in functional mode.

Table 6-7. RAMERRSTATUS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	RESERVED	R	0h	Reads return 0. Writes have no effect.
2	ADDR_DEC_FAIL	R/W1CP	0h	Address decode failed. This bit indicates that an address error interrupt was generated by the redundant address decode and compare logic due to a functional failure. This bit must be cleared by writing 1 to it in order to enable the capture of uncorrectable error address for subsequent failures. This bit has to be in a cleared state for generation of a new address error interrupt.
1	RESERVED	R	0h	Reads return 0. Writes have no effect.
0	SERR	R/W1CP	0h	Single Error Status. This bit indicates that the single-bit error threshold has been reached. This bit is set even if the single-bit error threshold interrupt is disabled. This bit must be cleared by writing 1 to it in order to clear the interrupt request and to enable subsequent single-bit error interrupt generation.

6.7.6 RAMSERRADDR Register (Offset = 14h) [reset = 0h]

RAMSERRADDR is shown in Figure 6-8 and described in Table 6-8.

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RAMSERRADDR captures the address for which the Cortex-R4F CPU detected a single-bit error.
The SERR bit in the RAMERRSTATUS register must be cleared, by writing 1 to the bit, in order to enable the RAMSERRADDR register to capture a subsequent new error address.

Figure 6-8. RAMSERRADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED														SINGLE_ERROR_ADDRESS	
R-0h														R-0h	
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
SINGLE_ERROR_ADDRESS													RESERVED		
R-0h													R-0h		

Table 6-8. RAMSERRADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-18	RESERVED	R	0h	Reads return 0. Writes have no effect.
17-3	SINGLE_ERROR_ADDRESS	R	0h	This register captures bits 17-3 of the address for which the Cortex-R4F CPU detects a single-bit error when the RAMTHRESHOLD register is set to 1. The lower 3 bits are always tied to zero so that the address captured is a double-word (64-bit) address. This is a 64-bit-aligned address is stored as an offset from the base of the TCRAM or ECC memory. This register can only be reset by asserting power-on reset, and holds the last error address even after a system reset.
2-0	RESERVED	R	0h	Reads return 0. Writes have no effect.

6.7.7 RAMUERRADDR Register (Offset = 1Ch) [reset = 0h]

RAMUERRADDR is shown in Figure 6-9 and described in Table 6-9.

Return to Summary Table.

RAMUERRADDR captures the address for which the Cortex-R4F CPU detected a multi-bit error.

Figure 6-9. RAMUERRADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED									UNCORRECTABLE_ERROR_ADDRESS						
R-0h									R-0h						
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
UNCORRECTABLE_ERROR_ADDRESS													RESERVED		
R-0h													R-0h		

Table 6-9. RAMUERRADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-23	RESERVED	R	0h	Reads return 0. Writes have no effect.
22-3	UNCORRECTABLE_ERROR_ADDRESS	R	0h	This register captures the address for which there was an uncorrectable error. The uncorrectable error is indicated by the Cortex-R4F CPU's SECDED logic. For the SECDED multi-bit or double-bit uncorrectable error this register stores bits 17-3 of the TCM access address. The lower 3 bits 2-0 are always read as zeros to indicate that the latched address is a double-word address. The address bits 31-18 are read as zeros. This is a 64-bit-aligned address is stored as an offset from the base of the TCRAM or ECC memory. For a redundant address decode and compare logic error this register stores the complete TCM access address rounded to a double-word boundary (bits 22-3). This error is also indicated by the ADDR_DEC_FAIL flag in the RAMERRSTATUS register. No error address is stored as a result of a redundant address logic test. The register has to be read-cleared to enable further error address captures. Reading the register does not clear its contents but enables the register to be updated with an uncorrectable error address. This register can only be reset by asserting power-on reset, and holds the last error address even after a system reset.
2-0	RESERVED	R	0h	Reads return 0. Writes have no effect.

6.7.8 RAMTEST Register (Offset = 30h) [reset = 5h]

RAMTEST is shown in Figure 6-10 and described in Table 6-10.

Return to Summary Table.

RAMTEST controls the test mode of the TCRAM Module.
For equality test:
If the comparator matches (no true silicon fail), there is no status bit set for ADDR_COMP_LOGIC_FAIL or ADDR_DEC_FAIL.
If there is true silicon malfunction, ADDR_COMP_LOGIC_FAIL and ADDR_DEC_FAIL will be set, no UERRADDRESS is captured.
For inequality test, the compare vector will not match since non-inverted and inverted values of the same test vector are fed to the comparator:
If there is no silicon malfunction on any of the comparator bits, then only ADDR_DEC_FAIL will be set. This is chosen so that we can ensure the functional ADDR_DEC_FAIL status bit data path can be tested.
If there is a silicon malfunction on any of the comparator bits, then, ADDR_COMP_LOGIC_FAIL and ADDR_DEC_FAIL will be set, no UERRADDRESS is captured.

Figure 6-10. RAMTEST Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							TRIGGER
R-0h							R/WP-0h
7	6	5	4	3	2	1	0
TEST_MODE		RESERVED		TEST_ENABLE			
R/WP-0h		R-0h		R/WP-5h			

Table 6-10. RAMTEST Register Field Descriptions

Bit	Field	Type	Reset	Description
31-9	RESERVED	R	0h	Reads return 0. Writes have no effect.
8	TRIGGER	R/WP	0h	Test Trigger. This is an auto reset test trigger used to test the redundant address decode and compare logic. A redundant address decode test is executed when test mode is enabled and the test trigger is applied by writing a 1 to this bit. The trigger is valid only if test is enabled and the correct mode is configured in the TEST_MODE field, and the ADDR_DEC_FAIL, ADDR_COMP_LOGIC_FAIL, and DERR flags are cleared in the RAMERRSTATUS register and the RAMUERRADDR register is read-cleared.

Table 6-10. RAMTEST Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-6	TEST_MODE	R/WP	0h	Test Mode. This field selects either equality or inequality testing schemes. 0h = Reserved 1h = Inequality check is done. The test stimulus stored in ADDRTEST_VECT register is inverted and fed into one channel and the non-inverted vector is fed into the other channel. If the XOR of these inputs is zero, then the UERR interrupt is generated and ADDR_COMP_LOGIC_FAIL flag is set in RAMERRSTATUS register. 2h = Equality check is done. The test stimulus stored in ADDRTEST_VECT register is fed directly to both the channels of the comparator. If the XOR of these two inputs is not zero, then UERR interrupt is generated and ADDR_COMP_LOGIC_FAIL flag is set in RAMERRSTATUS register. 3h = Reserved
5-4	RESERVED	R	0h	Reads return 0. Writes have no effect.
3-0	TEST_ENABLE	R/WP	5h	Test Enable. This is a 4-bit key to enable the redundant address decode and compare logic test scheme. If the test scheme is enabled, then the compare logic uses the test vector inputs from the ADDRTEST_VECT register. The functional path comparison is disabled when test mode is enabled. All other values = Test mode is disabled. Ah = Test mode is enabled.

6.7.9 RAMADDRDECVECT Register (Offset = 38h) [reset = 0h]

RAMADDRDECVECT is shown in Figure 6-11 and described in Table 6-11.

Return to Summary Table.

RAMADDRDECVECT is used for testing the redundant address decode and compare logic of the TCRAM Module.

Figure 6-11. RAMADDRDECVECT Register

31	30	29	28	27	26	25	24
RESERVED					ECC_SELECT	RESERVED	
R-0h					R/WP-0h	R-0h	
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RAM_CHIP_SELECT							
R/WP-0h							
7	6	5	4	3	2	1	0
RAM_CHIP_SELECT							
R/WP-0h							

Table 6-11. RAMADDRDECVECT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-27	RESERVED	R	0h	Reads return 0. Writes have no effect.
26	ECC_SELECT	R/WP	0h	ECC Select. This bit is used to store the ECC select value for the redundant address decode and compare logic. The stored value is passed as test stimulus for the built-in test scheme.
25-16	RESERVED	R	0h	Reads return 0. Writes have no effect.
15-0	RAM_CHIP_SELECT	R/WP	0h	RAM Chip Select. This field is used to store the RAM chip select value for the redundant address decode and compare logic. The stored value is passed as test stimulus for the built-in test scheme.

6.7.10 RAMPERRADDR Register (Offset = 3Ch) [reset = X]

RAMPERRADDR is shown in Figure 6-12 and described in Table 6-12.

Return to Summary Table.

RAMPERRADDR stores the address for which an address-parity error was detected.

Figure 6-12. RAMPERRADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED									ADDRESS_PARITY_ERROR_ADDRESS						
R-0h									R-X						

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDRESS_PARITY_ERROR_ADDRESS													RESERVED		
R-X													R-0h		

Table 6-12. RAMPERRADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-23	RESERVED	R	0h	Reads return 0. Writes have no effect.
22-3	ADDRESS_PARITY_ERROR_ADDRESS	R	X	Parity Error Address. This register stores the double-word boundary (bits 22-3) of the TCM access address for which there was an address parity error. This register must be read-cleared to enable further error address captures. Reading the register does not clear the register contents but enables the register to be updated with a new parity error address. This is a 64-bit aligned address stored as an offset from the base of the TCRAM or ECC memory.
2-0	RESERVED	R	0h	Reads return 0. Writes have no effect.

6.7.11 INIT_DOMAIN Register (Offset = 40h) [reset = FFh]

INIT_DOMAIN is shown in Figure 6-13 and described in Table 6-13.

Return to Summary Table.

INIT_DOMAIN is not supposed to be written into when a memory initialization is proceeding. In other words, the INIT_DOMAIN register has to be programmed prior to the generation of the SYS_TCRAMW_MMI_INIT_I pulse from the system module. The INIT_DOMAIN register has to be programmed based on the power domains that are awake and the hardware shall not provide any indication if a powered down domain is selected for auto-memory initialization.

Figure 6-13. INIT_DOMAIN Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								AUTO_MEM_INIT_ENABLE							
R-0h								R/WP-FFh							

Table 6-13. INIT_DOMAIN Register Field Descriptions

Bit	Field	Type	Reset	Description
31-8	RESERVED	R	0h	Reads return 0. Writes have no effect.
7-0	AUTO_MEM_INIT_ENABLE	R/WP	FFh	These bits are used to enable auto-memory initialization per power domain. These bits are all 1 by default and need to be written to 0 to prevent a certain power domain from getting initialized. Note that these bits are only enable bits for their respective power domains and the indication for the start of Memory Initialization would still come from the system module through the pulsed input signal SYS_TCRAMW_MMI_INIT_I. bit[0] represents the enable bit for power domain 0. bit[1] represents the enable bit for power domain 1. bit[2] represents the enable bit for power domain 2. bit[3] represents the enable bit for power domain 3. bit[4] represents the enable bit for power domain 4. bit[5] represents the enable bit for power domain 5. bit[6] represents the enable bit for power domain 6. bit[7] represents the enable bit for power domain 7.